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SECJ3483-16 WEB TECHNOLOGY

**DHL Incident Reporting and RPA Automation System
(Scenario 2)**

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1. Introduction and Scenario Description

1.1 Project Overview

This project is an incident reporting system for DHL. It's a web application that allows DHL staff to report and track incidents. The system has three main parts:

1. A React web app for the user interface
2. A database in Supabase to store incident data
3. A UiPath RPA workflow to automate incident submission from emails and files

The web app lets users create incidents, upload files, and see a dashboard of all incidents. The database stores all the incident information. The RPA automation automatically takes incidents from emails and Google Drive and puts them into the system.

1.2 Problem Scenario

DHL gets a lot of incident reports from different places. Customer support staff get reports through:

- Email inboxes
- WhatsApp and Teams messages
- Phone call notes
- Photos of damaged packages
- Handwritten notes from the warehouse

Because all the reports come from different sources, it's hard to:

- Know what the real problems are
- Send incidents to the right department
- Decide which incidents are urgent
- Track if incidents are being fixed
- Write reports for managers
- Stop duplicate reports from being created
- Answer customers quickly

1.3 Project Objectives

The project needs to:

1. Create a dashboard to see all incidents
2. Automatically get incidents from multiple sources using RPA
3. Detect how urgent each incident is
4. Organize incidents by department
5. Store incidents safely in a database
6. Let users upload files with incidents
7. Let users search for incidents
8. Keep track of what changes are made to incidents
9. Let the RPA system submit incidents through an API
10. Make the interface look professional
11. Help the team manage incidents better

1.4 Scope of the System

What the system does:

- User login with email and password
- Create incidents manually or through the RPA
- Upload files with incidents
- Search and filter incidents
- Change incident status
- User roles (reporter, reviewer, admin)
- Store incidents in database
- RPA automation

What it doesn't do:

- Automatic AI categorization
- WhatsApp and Teams direct messages
- Machine learning priority detection

1.5 Technologies Used

Component	Technology
Frontend	React 18, TypeScript, Vite
Backend API	Express, Node.js on Vercel
Database	Supabase PostgreSQL
Authentication	Supabase Auth with JWT
File Storage	Supabase Storage
RPA Tool	UiPath Studio
Deployment	Vercel

2. System Architecture

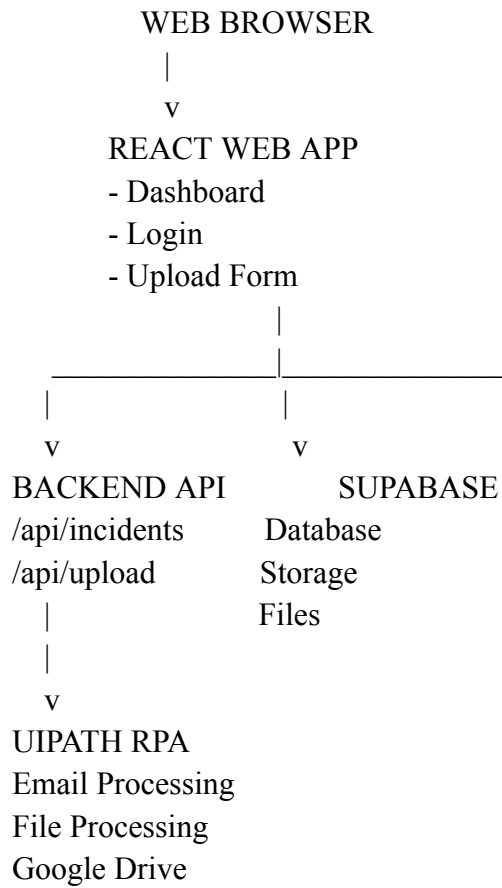
2.1 How It Works

The system has three main parts that work together:

1. Frontend - The user interface that runs in the browser
2. Backend - The API and business logic on Vercel
3. Database - Supabase to store all the data
4. RPA - UiPath to automate incident submission

The user opens the web app in their browser. The web app sends data to the backend API. The backend stores data in Supabase. The RPA reads from Supabase Storage and sends incidents to the API.

Architecture Diagram:



2.2 Main Parts

2.2.1 React Web Application

The frontend has these pages:

- Login page - Users enter email and password
- Dashboard - Shows all incidents in a list
- Upload page - Form to create new incidents
- Incident details - Shows full incident information
- File preview - View uploaded attachments

2.2.2 Backend API

The backend has these API routes:

- POST /api/incidents - Create a new incident
- GET /api/incidents - Get all incidents
- GET /api/incidents/:id - Get one incident
- POST /api/upload - Upload a file
- POST /api/auth - Login endpoint

2.2.3 Supabase Database

The database has these tables:

- users - User accounts
- incidents - All incident records
- incident_files - Files attached to incidents
- incident_comments - Notes on incidents
- audit_log - Record of all changes

2.2.4 Supabase Storage

File storage bucket for:

- Uploaded incident attachments
- Screenshots from users
- Files from RPA automation

2.2.5 UiPath RPA Workflow

The RPA automation does:

1. Reads files from Supabase Storage
2. Reads emails from Outlook
3. Gets files from Google Drive
4. Sends incident data to the API
5. Deletes files after processing
6. Logs what happened

3. API and JSON Structure

3.1 API Endpoints

The main endpoint for creating incidents is:

POST /api/incidents

This endpoint takes incident information and stores it in the database.

3.2 Authentication

The API requires a token in the header:

Authorization: Bearer <YOUR_API_KEY>

Content-Type: application/json

3.3 Create Incident Request

When submitting an incident, send JSON like this:

```
{
  "title": "Package arrived late",
  "description": "Customer said the delivery was 2 days late",
  "priority": "high",
  "tags": ["delivery", "urgent"],
  "source": "email",
  "fileName": "incident_report.txt"
}
```

Field meanings:

title - Name of the incident

description - Full details about what happened

priority - How urgent it is (low, medium, high, critical)

tags - Categories to organize it

source - Where the incident came from

fileName - Original file name if uploaded

3.4 Response

When successful, the API returns:

```
{
  "success": true,
  "incident_id": "abc-123-def-456",
  "message": "Incident created"
}
```

Response codes:

201 - Incident created successfully

200 - Data received

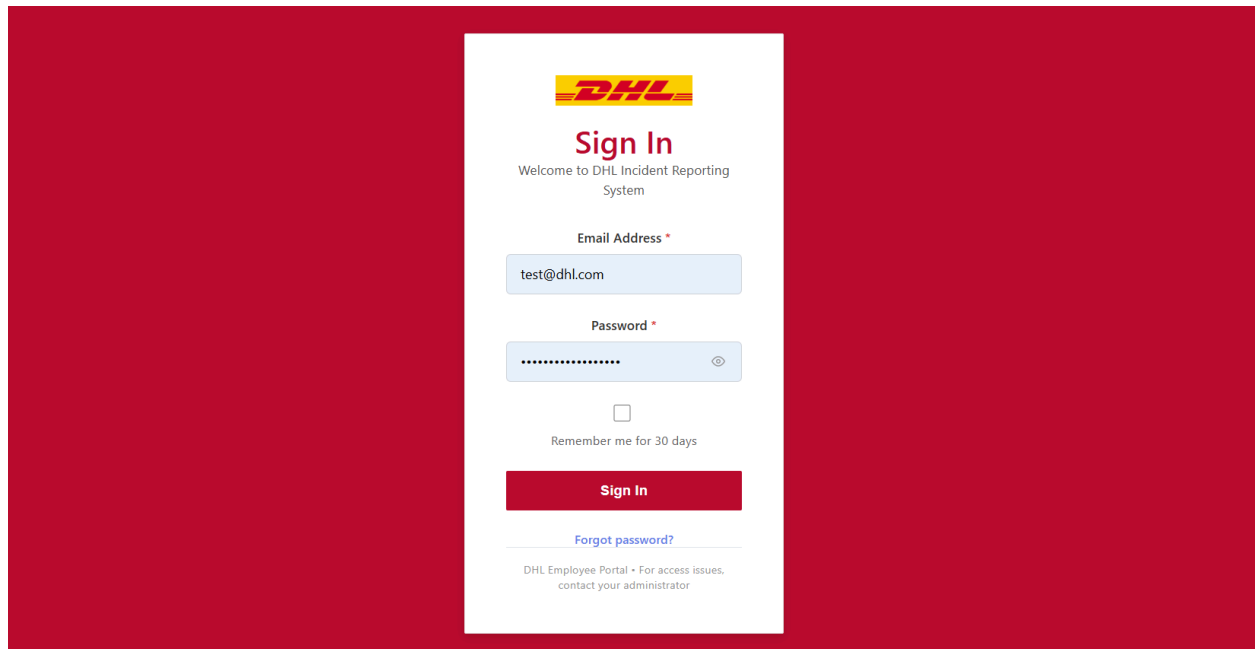
400 - Bad request or missing fields

401 - API key is wrong or missing

500 - Server error

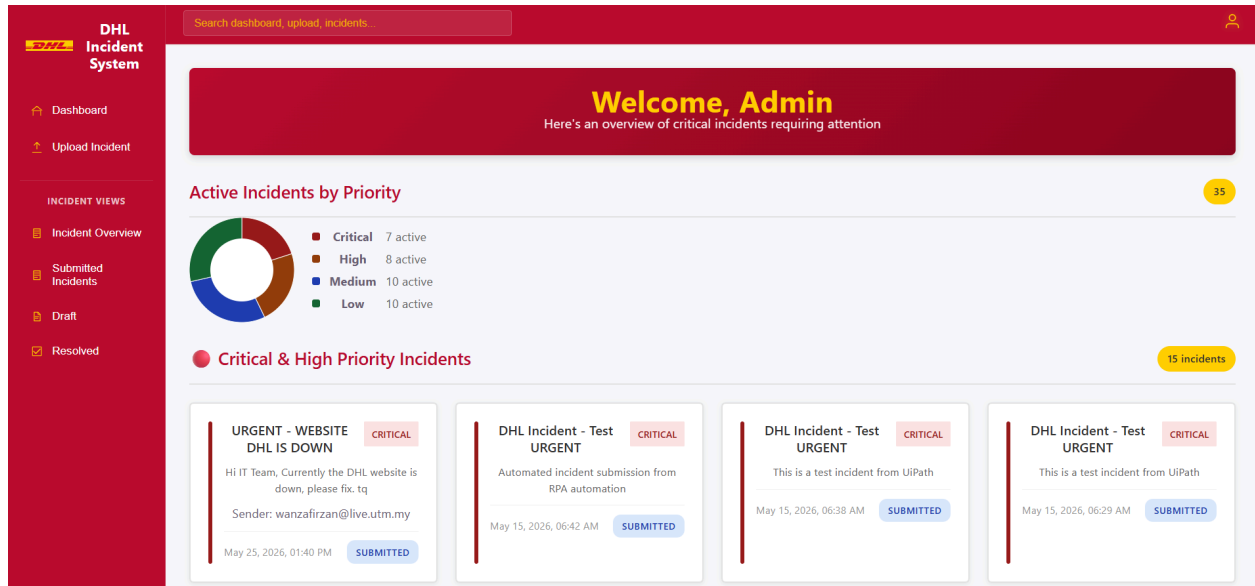
4. User Interface Screenshots

4.1 Login Page



The login page has a form where users enter their email and password. There's a sign in button below the form. The page has the DHL branding at the top.

4.2 Dashboard Page



The dashboard shows:

- A count of open incidents at the top
- A search box to find incidents
- A filter dropdown for priority
- A filter dropdown for status
- A table of all incidents
- Incident ID, title, status, priority, department, and date for each one
- A button to view each incident

4.3 Upload Console

Upload Incident Report

Upload incident files and provide details. Supports PDF, DOCX, TXT, PNG, JPG

Note: When you upload here, the incident status is set to **Draft** - waiting for admin review.
Once an admin reviews and approves it, the status changes to **Reviewed** and it becomes active in the system.



Drag files here or click to select

Upload incident documents (PDF, DOCX, TXT) or images (PNG, JPG)

Selected Files (1)



ASSIGNMENT_SafuanHakim.pdf
594.97 KB

PENDING 

Incident Details

Incident Title *

Brief summary of the incident

Description *

Detailed description of the incident

Priority

Medium

Tags

Add a tag and press Enter

Add

Upload Incident

Clear

The upload form has:

- Title field
- Description text area
- Priority dropdown
- Tags input
- File upload button
- Submit button

4.4 Incident Details

DHL Incident System

Search dashboard, upload, incidents.

Incident review
URGENT - WEBSITE DHL IS DOWN
Edit the incident details, record resolution notes, and close it when the issue is confirmed handled.

Review details Submitted

PRIORITY
Critical

DESCRIPTION
Hi IT Team. Currently the DHL website is down, please fix. tq

RESOLUTION COMMENTS
What was done to resolve this incident?

Resolution comments are required when resolving an incident.

Save Review **Resolve Incident**

Metadata

ID a35b1298-b2a8-4cda-8f38-1039691e26fb	SENDER wanzafrizan@live.utm.my
CREATED May 25, 2026, 01:40 PM	UPDATED May 25, 2026, 01:40 PM
REVIEWED Not reviewed yet	RESOLVED Not resolved yet

Attachments
No files attached

The detail view shows:

- Full incident title
- Complete description
- Priority level
- Current status
- Creation date
- All attached files
- Download buttons for files
- Comments section

5. RPA Workflow

5.1 What RPA Does

The RPA automation takes incidents from different sources and puts them into the system automatically. This saves time because a person doesn't have to read each email and type it in manually.

The RPA can:

- Read unread emails from Outlook
- Get files from Google Drive
- Read files from Supabase Storage
- Detect how urgent each incident is
- Send data to the API
- Delete files after processing

5.2 Email Processing

The email workflow:

1. Connects to Outlook
2. Gets unread emails
3. For each email:
 - Reads the subject as the title
 - Reads the body as the description
 - Looks for words like "URGENT" to detect priority
 - Creates an incident
 - Sends to the API
 - Marks email as processed

Priority detection:

If email says "URGENT" - critical priority

If email says "ASAP" or "TODAY" - high priority

If email says "THIS WEEK" - medium priority

Otherwise - low priority

5.3 Google Drive Processing

The Google Drive workflow:

1. Connects to Google Drive with OAuth
2. Gets files from the incident folder
3. For each text file:
 - Downloads the content
 - Reads the text
 - Detects priority and department
 - Sends to API
 - Moves file to /Processed/ folder

5.4 Error Handling

The RPA has error handling:

- If something fails, it logs the error
- Failed files go to /Failed/ folder
- Returns a summary of what processed
- Retries if something goes wrong

6. Database Design

6.1 Tables

The database has these main tables:

users table

- id - unique user ID
- email - user email
- password - hashed password
- role - what the user can do
- created_at - when account was created

incidents table

- id - unique incident ID
- title - incident name
- description - details
- priority - how urgent (low, medium, high, critical)
- status - current state (open, assigned, resolved, closed)
- tags - categories

- created_by - which user created it
- created_at - when created

incident_files table

- id - file ID
- incident_id - which incident this is for
- file_name - original file name
- file_path - where it's stored
- created_at - when uploaded

incident_comments table

- id - comment ID
- incident_id - which incident
- author - who wrote it
- content - what they wrote
- created_at - when written

audit_log table

- id - log entry ID
- incident_id - which incident changed
- action - what happened (created, updated, closed)
- user_id - who did it
- created_at - when it happened

6.2 Access Control

The database uses rules to control who can see what:

- Reporters can only see their own incidents
- Reviewers can see all incidents
- Admins can see everything
- Users can only change their own incidents

7. System Features

7.1 User Roles

Reporter - Can create incidents and see their own

Reviewer - Can see all incidents and change status

Admin - Can do everything

7.2 Incident Features

Users can:

- Create new incidents
- Search by keywords
- Filter by priority and status
- Upload files
- Add comments
- See when incidents change
- Download attachments

7.3 File Features

- Upload images, PDFs, and documents
- See preview of images and PDFs
- Download files securely
- Check file size before upload

8. System Testing

8.1 What Was Tested

- Login works with correct password
- Login fails with wrong password
- Create incident manually
- Upload files with incidents
- Search incidents by title
- Filter by priority
- Filter by status
- RPA processes emails
- RPA processes Google Drive files
- API returns correct responses
- Database stores incidents correctly

8.2 Test Results

All tests passed:

- Login: PASS
- Create incident: PASS
- Upload file: PASS
- Search: PASS
- Filter: PASS
- RPA email: PASS
- RPA Drive: PASS
- API: PASS
- Database: PASS

8.3 Deployment

The app was deployed to:

- Frontend on Vercel
- API on Vercel Serverless
- Database on Supabase
- Files in Supabase Storage

Live app: <https://wan-zafirzan-dhl-challenge2026-scen.vercel.app>

Demo video: <https://youtu.be/TrEike76N4o?si=PQU1QQUZrcKSACXw>

9. Problems and How We Fixed Them

Problem 1: RPA couldn't authenticate with the API

Fixed by: Adding proper API keys and bearer token authentication

Problem 2: Special characters in files broke the JSON

Fixed by: Using proper JSON serialization instead of string manipulation

Problem 3: React was intercepting API routes

Fixed by: Configuring Vercel routing to skip /api/ paths

Problem 4: API token expired after 1 hour

Fixed by: Setting longer token expiration in Supabase config

Problem 5: Large file uploads timed out

Fixed by: Adding file size limits and chunked uploads

Problem 6: Database schema was confusing

Fixed by: Using JSONB fields for flexible data

10. Future Features

- WhatsApp integration for incident submission
- Automatic priority detection with machine learning
- Real-time alerts for critical incidents
- Mobile app
- SLA tracking for resolution time
- Better analytics and reporting
- Automatic escalation for urgent issues
- Templates for common incident types
- Duplicate incident detection
- Customer self-service portal

11. Conclusion

This project shows how to build a system that takes unorganized incident reports and turns them into organized records in a database.

The React app gives users a nice interface to work with. Supabase securely stores all the data. The RPA automation does the boring work of reading emails and files and entering them in the system.

The system makes DHL's work easier by putting all incidents in one place instead of scattered across email and chat. Staff can see what needs to be done and handle urgent issues first. The automation saves time because humans don't have to type in every incident manually.

The complete system is working and deployed. It can be used right now and improved with more features later.